

MIAMA health outcomes: calculation and plot aggregation

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Reading guide

This extract covers health calculation and aggregation from the MIAMA-HUB README. Read **Health outcome units and aggregation** for the meanings of plot values: expected deaths/cases, life-years/HALYs, absolute benefit, percent improvement, per-100,000 denominators, time aggregation and mode attribution.

Documentation source: /private/tmp/MIAMA-HUB-release.

Counterfactual health outcomes

`apply_counterfactual_health_outcomes()` is the Step 7 bridge from changed counterfactual physical activity to health model outcomes. It uses the updated MIAMA-HM death-share cycle artifacts:

- `sp_cycle_outcomes_death_share`
- `mmet_d_cycle_lookup_death_share`

Those artifacts are produced by `MIAMA-HM/scripts/sp_hm_join.R`. HUB does not source or duplicate that script; it reads the processed artifacts and reproduces the scenario lookup-application logic documented in `MIAMA-HM/scenario_verification/scen_30to45_2mmets.qmd`.

Terminology in this step is explicit:

- **ref**: without-scheme/reference scenario
- **cf**: with-scheme/counterfactual scenario
- **delta**: `cf - ref`
- **baseline**: avoided here except when referring to HM source tables, because it can also mean the first simulation year

The function keeps all filtered individuals and all HM cycle rows. For the current `scheme_effect_duration = "longterm"` setting, the individual-level MMET delta from Step 6 is applied to every model cycle:

```
mmets_delta = mmets_cf_ind - mmets_ref
mmets_new   = mmets_cycle + mmets_delta
```

It then caps MMETs to the lookup maximum, overlaps the changed MMET interval with lookup bands, multiplies overlap width by per-MMET outcome slopes, and adds `d_*` outcome columns. Counterfactual values are derived as `ref + delta`; redundant `*_cf` columns are not materialized by default.

After applying the lookup, HUB also calculates health-adjusted life years (HALYs). This is distinct from the existing healthy-life-year (HLY) stream. Following the MIAMA-HM scenario-verification method, HUB reconstructs annual disease prevalence for each person. For chronic diseases:

```
prevalence[t] = prevalence[t-1] + incidence[t]
               - deaths[t] * death_share[t]
```

For depression, the remission/exits stream replaces the death-share term. Reference and counterfactual prevalence are calculated separately and divided by the corresponding population alive in each cycle. HUB then applies the packaged age/sex residual-disability (`pyld_rate`) table and age/sex/disease comorbidity-adjusted disability weights (`dw_adj`):

```
HALY = LY * (1 - pyld_rate) * (1 - sum(prevalence * dw_adj))
```

When no population remains alive in a person-cycle, prevalence disability and HALYs are set to zero. Person-cycles above the maximum age represented in the HALY parameter tables are retained in the health data but have NA HALY fields, so they are excluded from HALY summaries without discarding other outcomes.

The resulting person-cycle fields are `haly` (reference), `d_haly` (`cf - ref`), and, when `include_cf_columns = TRUE`, `haly_cf`. The two small parameter tables are packaged under `inst/extdata/data/health_data/haly_parameters`; their paths and the included disease streams are explicit in `cfg$results$haly`. The counterfactual health report records whether HALYs were available, the method, diseases used, and the unscaled total delta.

The death-share MMET lookup is much larger in memory than its parquet file. HUB derives the required (`age1year`, `female`, `mr_decile`, `cycle`) scope from the filtered people and cycle outcomes, applies Arrow filters before collection, and removes residual Cartesian combinations in R. Sample-mode calculations therefore do not materialize the complete England lookup table.

The low-level function returns `counterfactual_data` with:

- `health_outcomes`: full cycle-level reference and delta outcome table
- `counterfactual_health_report`: counts, MMET-delta summary, terminology, and summed outcome deltas
- `counterfactual_health_report$impact_overview`: compact outcome-level totals with `ref_total`, `cf_total`, `delta_total`, and `delta_per_1000_people`

Pass `include_cf_columns = TRUE` only for targeted model debugging that requires explicit `*_cf` columns. The high-level `Hub$build_results()` method consumes and releases `health_outcomes` after constructing compact `health_impacts` and `results_data`; plots and exports do not require the raw cycle table afterward.

Future work: add `scheme_effect_duration = "shortterm"`.

6.1 Health outcome units and aggregation

Start here for “what do the health numbers mean?” The input is not a table of observed diagnoses. HM simulates replicants, counts events and divides by the number of replicants. Its source quantities are per-person expectations. HUB matches/interpolates these to synthetic people and calculates CF changes. Summing person-level expectations gives expected counts, which may be fractional.

The authoritative source definitions are in MIAMA-HM `scripts/aggregation.R` (event counts divided by `n.rep`, but death shares are separate ratios), `scripts/sp_hm_join.R` (synthetic-person matching), and `scenario_verification/scen_30to45_2mmets.qmd` (scenario aggregation). HUB calculation details live in `R/results_data_prepare.R`, `R/results_data_plotting_functions.R` and `R/health_outcomes_calculate_halys.R`.

Source fields and their units

Field/family	Meaning in the source	Correct treatment in HUB
dead	Expected deaths per source person in a model cycle, normalized by original HM replicants	Sum across people/cycles for expected deaths; do not describe this as disease incidence
coronary_heart_disease, stroke, diabetes, depression, all_cause_dementia, individual cancer fields, parkinson.s_disease	Expected disease-event contributions per source person-cycle	Sum for expected disease cases/events, not the number of unique people with any disease
unhealthy	Net transition into diseased-or-dead state per source person-cycle	Accumulate to reconstruct healthy state; negative values can represent recovery/remission, not invalid incidence
depression_remission	Modelled depression remission/exits per person-cycle	Use in prevalence reconstruction; not a disease-benefit headline by itself
death_share_*	Proportion of deaths occurring with each disease, conditional on deaths in that cycle	Use in prevalence/HALY reconstruction; never sum as deaths or cases
ly, hly, yrs_* in other/overall HM artifacts	Life, healthy-life or disease-state years per modelled person	Not the current Tab 5 aggregation input; current LY/HLY are reconstructed from cycle states, avoiding summing pre-aggregated horizon totals again
haly, added by HUB	Health-adjusted person-years contributed in one model cycle	Sum across people and years; benefits have the opposite sign to prevented deaths/cases
d_*, added by lookup/HUB	CF minus REF, in the units of the corresponding field	Retain that sign internally; convert to benefit direction only for presentation

The UI catalogue maps mortality to **dead**; CVD to coronary heart disease plus stroke; cancers to the ten configured cancer streams; other options to their named underlying stream. The headline disease total counts each configured raw stream once. **Do not sum displayed CVD with IHD/stroke, or cancers with their displayed subtypes:** those selections overlap. Likewise deaths, cases and HALYs are different quantities and must not be added into one generic health total.

Aggregation formulas

Let $r[i, t]$ and $c[i, t]$ be REF/CF person-cycle contributions and w the configured person weight. The current model uses annual cycles.

```

annual REF expected total[t] = sum_i(w * r[i, t])
annual CF expected total[t]  = sum_i(w * c[i, t])
period total                 = sum over included positive model cycles
raw delta                    = CF total - REF total

deaths/cases prevented       = REF total - CF total
life-years/HALYs gained      = CF total - REF total
relative improvement (%)      = 100 * benefit / REF total
benefit per 100,000 people   = 100000 * benefit / represented population

```

Compute ratios **after summing their numerators and denominators**, not by averaging individual percentages. A zero REF denominator gives an unavailable percentage, not zero benefit. A period total per 100,000 is **not an incidence rate per person-year**; no at-risk person-time denominator is calculated here. Annual plots use represented people with source rows in that cycle. Cumulative plots and overall totals use the cohort

denominator (maximum population across cycles within each baseline age/sex stratum), not a sum of repeated person-years or a shrinking final-cycle denominator. Age groups use baseline age.

For modelled year contributions, transformation comes before aggregation:

```
LY[i,t] = 1 - cumulative_deaths[i,0:t]
HLY[i,t] = 1 - cumulative_net_unhealth[i,0:t]
HALY[i,t] = LY[i,t] * (1 - residual_pYLD[i,t])
           * (1 - sum_d(prevalence_among_alive[d,i,t] * adjusted_DW[d,i,t]))
```

HUB reconstructs prevalence from disease inflows and deaths/remission. See the HALY section above for details. Cycle 0 initializes these states but earns no reported life-year/HALY and is not counted as an incident event in Tab 5. This is a discrete annual-cycle convention, not a within-year survival integral. Currently main tables and plots include all available nonzero cycles, while AMAT/headline totals stop at the configured assessment horizon. This is not intrinsically wrong: it can represent full-follow-up plots alongside a shorter appraisal summary. It does require explicit period labels if those differ. Whether all products should instead use one horizon is an open decision; no change to that selection policy has been made in this review. Missing late-life source rows are not newly observed zero-event years.

One record, one appraisal person: user counts are real-person targets and health aggregation uses `person_weight = 1` by default for all datasets. Leeds' historical 163.552 source expansion is retained only as `source_person_weight`. Sampling, eligible donor pools and replacement rules are unchanged. Absolute Leeds health totals are consequently 1/163.552 of the previous weighted totals for identical sampled records; percentages and per-100,000 results are unchanged. Recreate `cfg` and rerun the appraisal: existing result objects and previously saved configurations retain their old weight. Explicit custom `person_weight` overrides remain supported but depart from the default real-person contract.

Labels and mode attribution

Use “expected deaths”, “expected disease cases”, “life-years gained” and “HALYs gained” as appropriate. For a mixed percentage overview use “health improvement from reference (%)”, since fewer deaths/cases but more HALYs are beneficial. Do not label the combined panel “incidence rates by disease”. Unlike outcome units must remain separate even when plotted together.

Mode-specific health effects are allocations of the combined person's health delta using signed mode-specific MMET changes. They are not separately modelled REF/CF burdens. Their absolute REF/CF values and relative percentages are therefore unavailable; attributable changes can be reported as counts or per 100,000. Opposing mode effects can produce negative allocations or shares above 100%, while the allocated effects sum to the combined effect.

Inspect actual source values

Run `inst/workflows/dev_health_outcome_audit.R` with HUB loaded. It writes the outcome catalogue, numeric field ranges/missingness and row counts by cycle. By default it reads the configured death-share cycle source. To inspect the calculated run instead, first set:

```
audit_health_data <- hub$counterfactual_data$health_outcomes
source("inst/workflows/dev_health_outcome_audit.R")
```

The 2026-09-07 Leeds source audit found 192,116 rows covering cycles 0-40: 5,000 rows in cycles 0-10, falling to 3,988 in cycle 40. `dead` ranged from 0 to 0.297, `diabetes` from 0 to 0.375, `unhealthy` from -0.067 to 0.789; death shares ranged within 0-1. No non-finite values occurred in these fields. These are descriptive checks, not independent validation of HM probabilities.

Remaining review points: aggregation currently uses `na.rm = TRUE` and can hide missing values in future datasets; custom outcome definitions must follow the supported event/year semantics; source follow-up coverage and the separate real/synthetic unit contract need continued review. The audit should be rerun when HM data are replaced.

`prepare_results_data()` prepares compact Step 8 outputs for MIAMA-UI. The expensive reference, counterfactual, and HM steps run before this function; interactive Tab 5 filters operate on compact result tables and do not rerun the health model.

Cycle 0 is the without-scheme baseline state and is excluded from presented impact totals and timelines. Cycle 1 is the first modelled year. Raw HM `d_*` columns retain the technical convention **counterfactual - reference**. Benefit-oriented fields use **reference - counterfactual** for adverse outcomes (deaths and disease cases) and **counterfactual - reference** for HALYs, so positive values consistently indicate a health gain. Legacy column names such as `prevented_value` are retained in the compact result contract even when the selected positive outcome is more naturally described as HALYs gained.

One selected appraisal person record represents one real person: `cfg$population$person_weight` defaults to 1 for sample, Leeds and full datasets. Entering 1,000 users still samples 1,000 records; the sampling pool, probabilities and feasibility rules are unchanged. Geographic expansion is retained only in `cfg$population$source_person_weight`, not applied to health totals or denominators. Trip counts remain row targets and totals without an extra expansion factor.

For identical records, absolute Leeds health totals are 1/163.552 of the previous weighted totals; relative effects and per-100,000 results are unchanged. Recreate the configuration and rerun the appraisal after updating HUB. Previously saved configurations/results retain their old weight. Explicit custom person-weight overrides remain supported but depart from the default real-person contract. Source-derived default counts describe an example appraisal, not the entire LAD; the full geographic population estimate in Tab 1 remains contextual information.

Headline health outcomes

After `build_results()`, UI can populate the four highlight figures with one method call:

```
highlights <- mdata[["hub"]]$get_results_highlights()
# The same table is returned directly as mdata[["result"]]$highlights.
# Stateless equivalent: get_results_highlights(mdata[["result"]]$results_data)
```

The returned rows are `premature_deaths_prevented`, `life_years_saved`, `halys_gained`, and `disease_cases_prevented`, with display labels, units, assessment period, and availability status. These totals intentionally ignore interactive Tab 5 filters. The disease total sums each underlying HM incidence stream once; presentation composites such as CVD/all cancers and their subtypes are not double-counted. It is a total of prevented disease events, not unique people.

Reporting units in the current UI

The age plot and timeline now pass `modes = NULL`: they show combined appraisal health outcomes filtered by age/sex/outcome. Mode selections apply to the separate mode-impact and trip plots only. The main overview remains fixed and unfiltered. Timeline Presentation affects only the timeline, not the age/mode bars.

Reporting unit	Benefit/difference plots	Separate REF/CF timeline
Absolute values	Deaths/cases prevented or HALYs gained	Expected deaths/cases or HALYs in each scenario
Values per 100,000 people	Benefit divided by assessed population, times 100,000	Each scenario value divided by assessed population, times 100,000
Percentage improvement	100 times benefit divided by REF outcome total	Unsupported; use the benefit view or another unit

Positive benefit means REF minus CF for deaths/cases and CF minus REF for HALYs. Age/mode bars cover the included assessment cycles; timelines are cumulative by default. Cumulative per-100,000 values use the represented cohort denominator, not a denominator shrinking as late-cycle source rows disappear. This is not an incidence rate per person-year. Outcome panels label their units explicitly and unlike outcomes are not added together. Benefit timelines display outcome lines on one shared scale; separate REF/CF timelines retain their outcome panels. Axes use short labels, with the period and detailed outcome units in subtitles, captions and tooltips. A shared scale does not make HALYs and event counts interchangeable or additive.

Mode-attributed percentages have no separate REF denominator. HUB now displays an explanatory unavailable view instead of silently switching to absolute values. Separate REF/CF plots likewise explain that percentage

improvement requires a difference view. UI captions come from the actual generated plot; filter-only changes do not rerun sampling or the health model.